USABILITY, WORKFLOW AND ACCESSIBILITY AUDIT

# MERIT Usability Audit

Can someone new to MERIT find where to start, act correctly at each step, and know when they have finished? Mostly yes. The places where they cannot are concentrated in a few high-traffic screens.

**Auditor:** Claude Code, Anthropic's AI coding assistant (model Claude Opus 5.5)

**Date:** Nov 2, 2026 **Build:** `main @ 8b03204` (PR #146), local demo database

**Type:** evaluation only; no code changes, no records saved

A. Summary | Method and limits | B. Workflows | C. Walkthroughs | D. Findings | E. Simplification | F. Roadmap | G. Scorecard | H. GitHub issues

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## A. Executive summary

**Overall:** MERIT is a well-built, consistent public-health application. Its navigation now groups features by job and shows each role only what it may use. Forms validate on the server with clear, focused error summaries. Automated accessibility testing found almost nothing. A trained user will be productive quickly.

**Where it falls short:** a first-time user will hesitate in a few dense screens, in the ones staff use every day. Most problems come from showing everything at once, or from a default that points the user the wrong way. Few come from missing features.

### Major strengths (verified)

- **Accessibility (automated):** axe-core reported **no violations** on 34 screens as the supervisor, the same as the administrator, and one as the county case manager, against WCAG 2.0, 2.1

### Major weaknesses

- **Work queues:** the daily task list is hard to read. Its most useful column is 116 px wide, so 15 tasks fill a 4,700 px page (UX-01, UX-07).
- **Held records:** the group correction

and 2.2 A and AA rules.

- **Reflow:** no page scrolls sideways at 768 or 390 px; at 1024 px and below the header folds behind a Menu button.
- **Navigation:** grouped and scoped to each role, and keyboard-operable (Enter opens, Escape closes and returns focus). Most screens mark their place in it.
- **Error recovery:** clear validation. Error summaries take focus and link to each field. When a session ends mid-form, the draft is kept.
- **Guidance:** a search-first pattern prevents duplicate people. First-time authenticator setup is detailed and works.

starts on *Last name* whatever the cause, inviting a wrong bulk correction (UX-02).

- **Case page:** it shows every form at once, including Override and a red Close case (UX-06).
- **Recovery codes:** they can be skipped by navigating away, and they are never shown again (UX-03).
- **Stored data:** a data defect puts lab units inside addresses in tasks (UX-04).

**Most important opportunities:**

1. Make the task list readable: put the detail under each task title, and use cards on phones.
2. Make every bulk correction start from the field that caused the hold, and name that field in the confirmation.
3. Put rarely used, high-stakes case actions behind a button.
4. Keep the recovery-code step from being skipped.

Items 2 and 4 are quick wins. Items 1 and 3 are medium-effort changes to screens used every day.

**Needs more testing:** real users (no usability sessions were run), screen readers, WAVE, real phones, and every workflow that saves data (see limits).

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# Method, evidence and limits

I used the running application from a real browser (Microsoft Edge through Playwright) as eight demo roles, signing each one in for the first time. I combined that with inspection of the source code.

### Roles tested

Supervisor, state administrator, county case manager (Baltimore City), intake clerk, epidemiologist, occupational investigator, city user administrator, and platform administrator.

### What I ran

- **First-time sign-in:** password, authenticator enrollment and recovery codes for all eight roles. The epidemiologist's run was captured step by step.
- **Navigation inventory:** the structure, tab order and header axe results for every role.
- **Page sweep:** 34 screens each as the supervisor, the administrator and the case manager. For each screen it recorded the heading, the current-location marker, back links, axe results (WCAG 2.0–2.2 A and AA), sideways scrolling at 1280, 768 and 390 px, console errors, and a full-page screenshot.
- **Eight task walkthroughs,** including a session-expiry test.
- **Keyboard checks:** the navigation groups, form fields and focus visibility.

### Evidence labels

Each finding is labelled with how it was established:

- **Verified** observed in the running application;
- **Code-inferred** read in the source code, not exercised;
- **Potential** plausible, and needs real-user testing.

Simulated first-time-user reasoning is **not** evidence from real users.

> ### Not evaluated
>
> - **Usability sessions:** none with real people.
> - **Accessibility tools:** no screen reader (NVDA, JAWS or VoiceOver) and no WAVE, because neither tool was available in this environment. So screen-reader compatibility, announcements, and WAVE's errors, alerts and contrast checks are untested.
> - **Zoom and devices:** 200% and 400% zoom were approximated with narrow viewports (640 and 320 px CSS) rather than browser zoom. Real phones and

tablets were not tested.

- **Workflows that save data:** to keep the audit non-destructive, I didn't save a person or result, upload a file, release, reject or correct held rows, decide a match, complete, assign or dismiss tasks, log case activity, override or close a case, request extracts, mark one submitted, create a user, tenant or profile, or use "view as". These were walked up to the last step before saving, or inspected.
- **Background processing:** the worker was not running, so I didn't observe background processing, such as an uploaded batch moving through validation.
- **Roles not signed in separately:** `cmbaltco`, `cmboth` and `countyadmin`. Their permissions match roles that were tested.

**Side effects (approved):** I redeployed the local MERIT demo database from `main`, and reset and re-enrolled MFA for the eight audited demo users. Views, searches and sign-ins were written to the audit log as usual.

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# B. Workflow inventory

| WORKFLOW AND PURPOSE | INTENDED ROLE | ENTRY POINT | PRIMARY STEPS | OUTCOME | CONCERNS |
|---|---|---|---|---|---|
| **First sign-in and MFA enrollment** | Everyone | /account/signin | Password → add MERIT to an authenticator (QR or key) → code → save recovery codes → continue | Signed in with MFA | Recovery codes can be skipped (UX-03) |
| **Daily triage** | Supervisor, case manager, epidemiologist | Home | Read tiles → *Needs attention* → open an item | Knows today's work | Duplicates, raw rule text (UX-05) |
| **Find a person** | Anyone with person.view | Header search, People | Search → results → person page | Record open | Look-alike records unflagged |

| | | | | | |
|---|---|---|---|---|---|
| | | | person page | | unflagged (UX-18) |
| **Add a person** | person.edit | People → search → Add a person | Search → no match → Add → form → Save | New record | Retyping, no h1 (UX-10) |
| **Enter a result** | result.edit | Person page → Enter a result | Analyte → specimen and unit cascade → value, dates, lab → Save | Result and case outcome | Unit spellings (UX-13) |
| **Upload a lab file** | Intake clerk | Intake → Intake | Choose sender profile → file → Upload → batch card | Batch processed | Jargon, truncated profile (UX-15) |
| **Fix and release held records** | Intake clerk, supervisor | Intake → Held records | Choose cause → correct a field → Release | Rows load | Field default (UX-02); next step shown only for matches |
| **Person match review** | Supervisor | Intake → Match review | Open record → compare candidates → Same, Different, New person or Not sure | Record linked | Repeated data, no primary action (UX-16) |
| **Work a queue task** | task.view, task.edit | Work queues | Choose queue → read task → open case or person → Complete, Assign or Dismiss | Task done | Unreadable detail (UX-01), phone (UX-07), empty link (UX-17) |
| **Manage a** | case.edit | Case page | Review → log | Case | All forms |

| | | | | | |
|---|---|---|---|---|---|
| **case** | | | activity, assign, override or close | updated | open (UX-06) |
| **Investigation** | case.edit, investigator | Surveillance → Investigations | Open → samples and findings → Save | Investigation updated | None found automatically |
| **Clusters and case map** | case.view | Surveillance → Clusters or Map | List → cluster → members, tasks and map | Cluster reviewed | None found automatically |
| **Run a report and export it** | result.view, case.view | Registry & reports → Reports | Choose report → dates and analyte → Apply → Export | Table, chart, file | Minor formatting (UX-14) |
| **Federal extract** | extract.run, extract.submit | Registry & reports → Extracts | Request → Download → Mark submitted | CDC or NIOSH file | Primary on Mark submitted; identical runs (UX-12) |
| **Manage users** | Administrators | Administration → Users | Add a user → Save → Add access | Account with access | Two-step access (UX-11) |
| **Audit log** | audit.view | Users → Audit log; record links | Filter → page → export | Who did what | Not in navigation for supervisors (UX-08) |
| **Recover from session expiry** | Everyone | Any form | Submit after the session ended → alert → sign | Work kept | Alert layout (UX-19) |

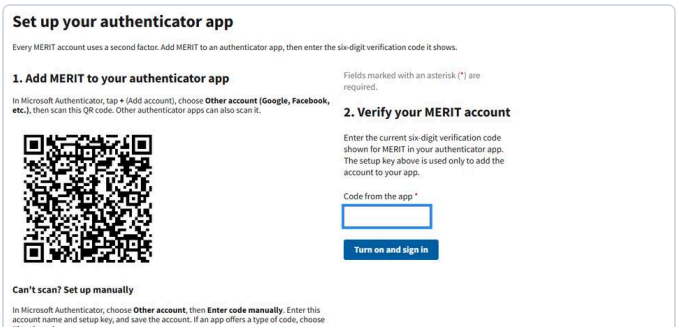
| | |
|---|---|
| | in in a new tab → Check session |

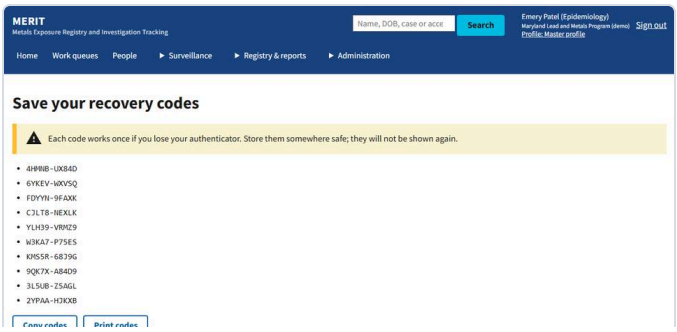
# C. Cognitive walkthrough results

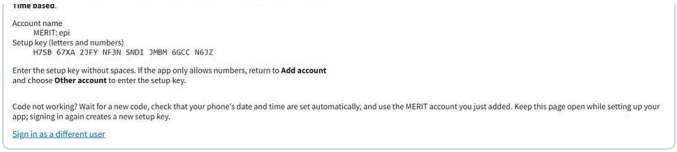
For each workflow I asked, at every step, the eight walkthrough questions: goal, action, control, action–result, feedback, progress, error recovery and next step. The tables keep the steps where an answer was *no* or *uncertain*. **Breaks at** names the first step where a first-time user is likely to stall or err.

## C1. First sign-in and authenticator setup Verified

| STEP | WHAT THE USER MUST DO | ASSESSMENT |
|---|---|---|
| 1. Sign-in | Enter user name and password, then Continue | Clear. The page says a code will follow. Focus starts in User name. |
| 2. Set up the authenticator | Scan the QR code, or enter the key manually, then type the code | Strong. Numbered steps, a manual fallback and troubleshooting text. The required-field note sits oddly between steps 1 and 2. |
| 3. Recovery codes | Save 10 codes, tick "I have saved these codes", Continue | **Breaks here.** The full menu and search are already live, so *Home* or a search leaves the step without saving. The page says the codes "will not be shown again" (UX-03). |
| 4. First landing | Understand Home | Tiles and *Needs attention* orient well, but repeated rows and rule jargon dilute them (UX-05). |






Authenticator setup: clear numbered steps and a manual fallback.




Recovery codes: the navigation and search are active before the codes are acknowledged.

## C2. Find a person and open a result Verified

Header search for "Rivera Ana" → 2 matches → person → result: **4 steps**. Every step is clear. The result page has a "Back to the person" link, but no navigation item is marked current, so the user's place is not shown (UX-08). In the results, two look-alike records (an "Ana Rivera" with no date of birth, and "Mia" and "Mya" Rivera with the same date of birth) carry no warning (UX-18, potential).

## C3. Add a person Verified

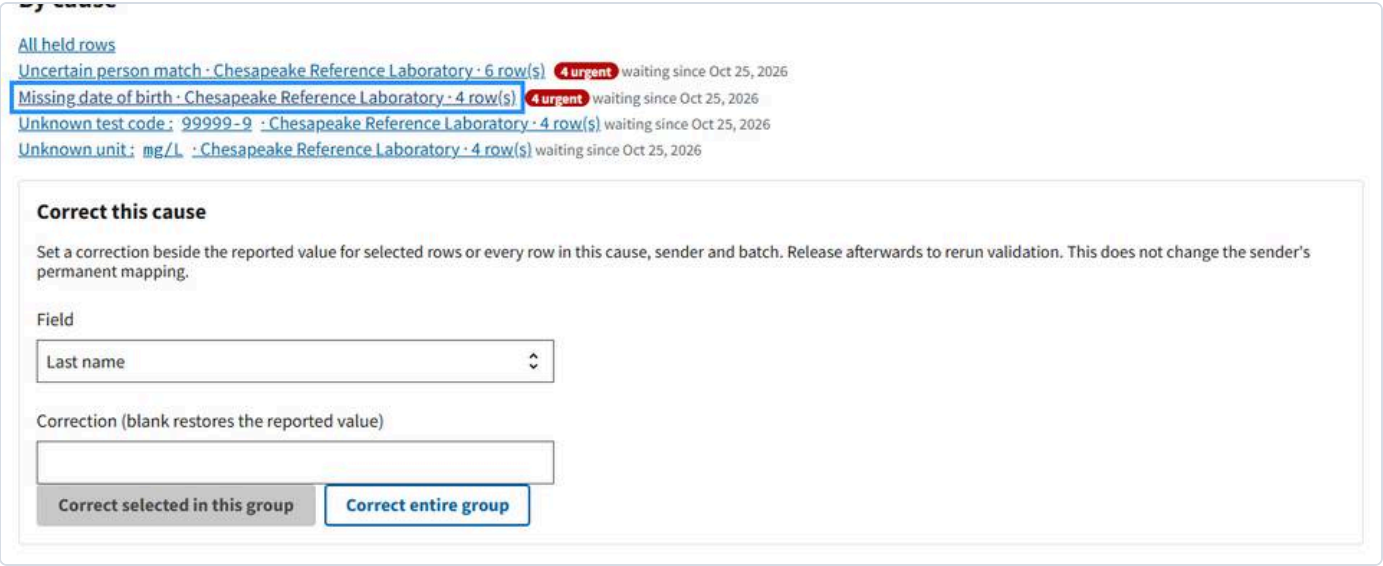
| STEP | WHAT THE USER MUST DO | ASSESSMENT |
|---|---|---|
| 1. Open People | Look for "Add a person" | Hesitation: the only button is Search. Adding appears only after a search, which is intentional, but nothing says so beyond "Search first...". |
| 2. Search, no match | Read "No matches in your access scope. Add a person" | Clear. |
| 3. Form | Retype the name and date of birth just searched | **Breaks here (efficiency).** Nothing is prefilled. The page has no h1 ("Add a person" is an h2) (UX-10). |
| 4. Empty Save | Read the errors | Strong: a focused error summary ("Last name is required.") with links to the fields. |

## C4. Enter a laboratory result Verified

Person → Enter a result → choose an analyte (the specimen and units cascade) → value, collection date, laboratory → Save. The collection date defaults to today, and validation lists exactly what is missing ("Result value is required. Laboratory is required."). Users hesitate at **Unit**: the list offers `ug/dL`, `mcg/dL` and `µg/dL`, three spellings of the same unit (UX-13).

## C5. Fix and release held records Verified up to the correction

| STEP | WHAT THE USER MUST DO | ASSESSMENT |
|---|---|---|
| 1. Held records | Understand why rows are held | Good: "By cause" groups the rows. "Fix the cause once, then release every row it held" states the model. |
| 2. Choose "Missing date of birth" | Read the *Correct this cause* panel | **Breaks here.** *Field* defaults to **Last name**. A user who types a date and presses **Correct entire group** writes it into Last name on every row. The confirmation ("Apply this correction to all 4 rows…") doesn't name the field (UX-02). |
| 3. Release | Select the rows → Release selected | A separate step. The next step is shown only for uncertain matches; other causes get no hint. |

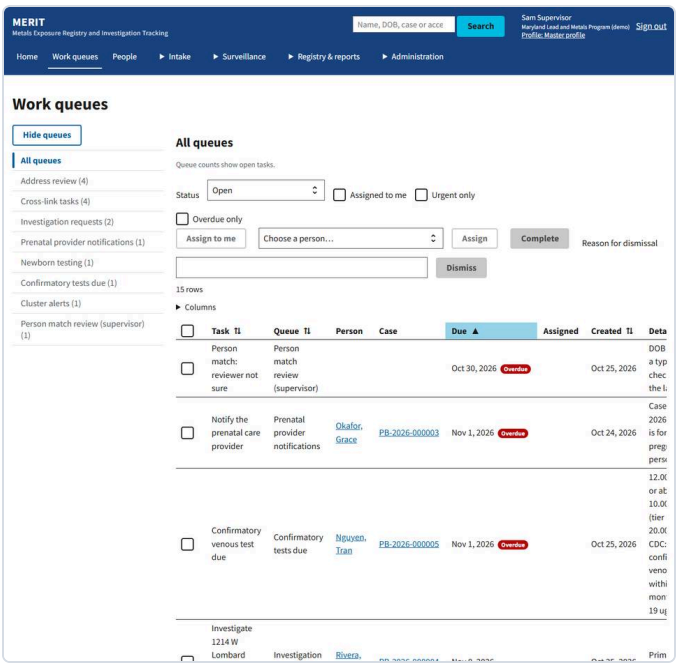



"Missing date of birth" is chosen, but the correction field starts on "Last name".

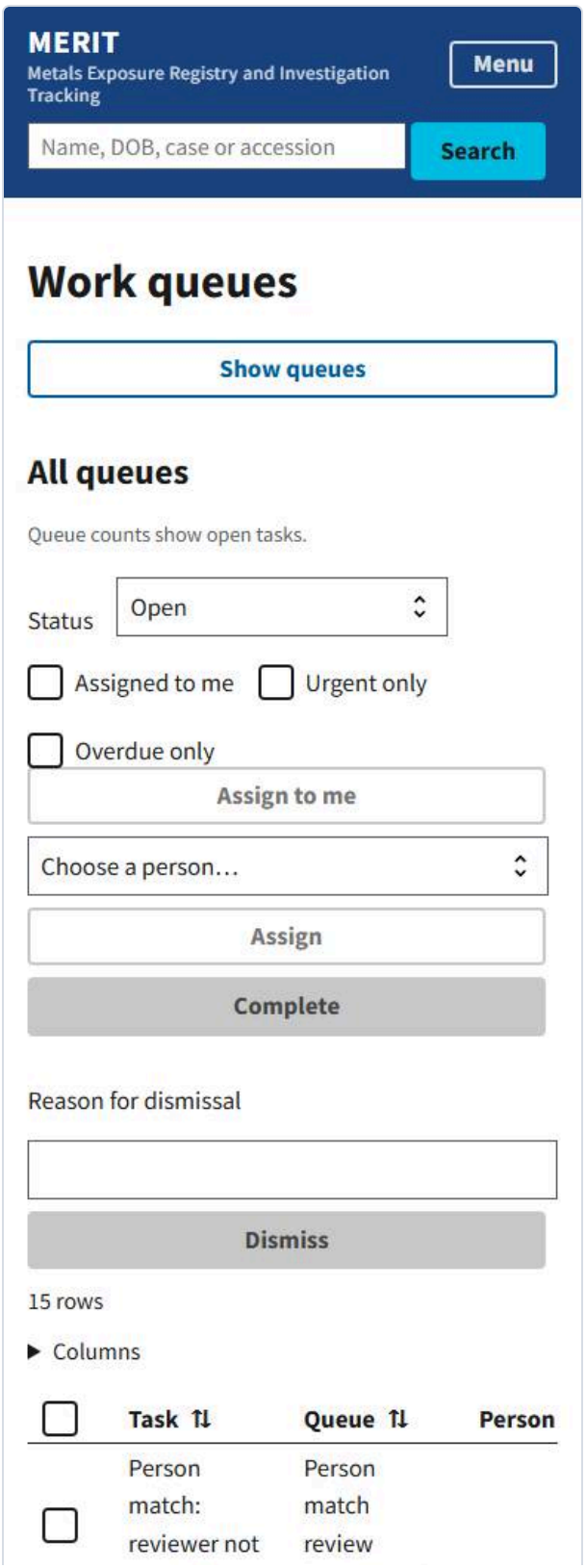
## C6. Work a queue task Verified up to selection

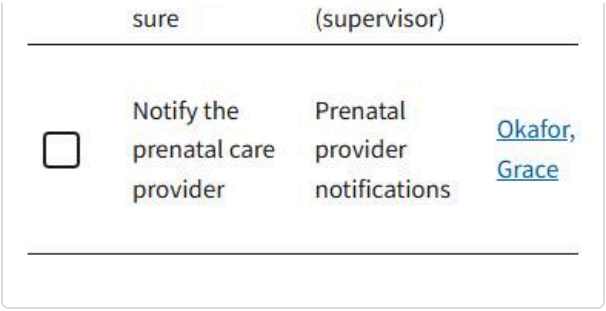
| STEP | WHAT THE USER MUST DO | ASSESSMENT |
|---|---|---|
| 1. Work queues | Find today's task | **Breaks here.** The Detail column, which explains each task, is 116 px wide and runs past the edge. Rows are 150–200 px tall, so 15 tasks make a 4,700 px page, and the grid needs sideways scrolling (UX-01). |
| 2. Select | Tick the row, | Clear: Assign to me, Complete and Dismiss become active. Dismiss |

| | | |
|---|---|---|
| a task | then choose an action | needs a reason, but its field sits on its own line away from the button. |
| 3. Act on it | Open the person or case first | Possible through the row's links. Some tasks show addresses with "ug/dL" in them (UX-04). |




1280 px: the Detail column is squeezed to a sliver at the right edge.



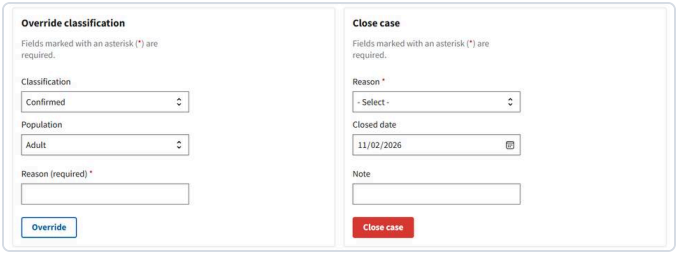



390 px: five stacked action controls come before the first task, and only three columns are visible (UX-07).

## C7. Manage a case Verified display

The case page puts the summary, open tasks with inline actions, results, the activity log, *Override classification* and a red **Close case** on one long page, all expanded. A first-time user meets the high-stakes forms beside routine work. The main *Save* button is a secondary outline, while *Log activity* is primary (UX-06).

When the session ended mid-entry, the activity note was kept, and the alert offered sign-in in a new tab plus *Check session*: good recovery. The two controls touch each other (UX-19).




Override and Close are always expanded.




Session ended: the draft is kept.

## C8. Submit a federal extract Verified display

Request → find the run → Download → Mark submitted. **Breaks at** choosing the run: three CBLS Q3 runs have identical titles and file names, with nothing marking the latest. **Mark submitted**, which records an external submission, is the primary blue button, while Download is secondary (UX-12).






Identical runs; the irreversible action is styled as the primary one.

### C9. Add a user Code-inferred

The Add a user form has no role or access fields. Save redirects to the user's edit page, where *Add access* is a separate form. A first-time administrator may stop after Save, leaving an account with no access. Nothing on the form says a second step follows (UX-11).

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# D. Detailed usability findings

20 findings: 2 high, 10 medium and 8 low. No critical issues: nothing stops a task from being completed. The severity reflects the impact on users and how often the workflow is used.

UX-01 High Verified

### Work queue task details are unreadable in the grid

| | |
|---|---|
| Affected | Work queues (Angular), task grid, every role with task.view |
| Problem | The Detail column, which explains why the task exists and what to do, is squeezed to 116 px and runs past the right edge of the grid. Each row grows to 150–200 px, so a page of 15 tasks is about 4,700 px tall. Task titles wrap word by word. |
| Evidence | At 1280 px: grid scrollWidth 1006 vs clientWidth 936; Detail header 116 px wide; page height 4,705 px (supervisor, demo data). Screenshot in C6. |
| Why users struggle | The user must scroll sideways, or open each task's case, to learn what the task asks. Scanning for the next job becomes reading a long, narrow column. |
| Recommendation | Show the detail as a second line under the task title (two lines, then "more"); give Task and Detail flexible widths and fix the others to their content; keep Detail out of the narrow column set. On phones, see UX-07. |

| | |
|---|---|
| Expected benefit | Read and choose tasks without scrolling sideways; about 3× more tasks on one screen. |
| Complexity | Medium |
| Heuristic / criterion | Nielsen 6 (recognition rather than recall), 8 (minimalist design) |

UX-02 **High** **Verified**

### Held-record group correction starts on the wrong field

| | |
|---|---|
| Affected | Held records → Correct this cause |
| Problem | Whatever the chosen cause (missing date of birth, unknown test code, unknown unit), the Field list starts on "Last name". Correct entire group applies the value to every row in the cause, sender and batch, and its confirmation ("Apply this correction to all 4 rows...") doesn't name the field. |
| Evidence | Observed for "Missing date of birth": Field = Last name. Code: the select lists the file fields in fixed order with no selection tied to the cause (held.page.ts). |
| Why users struggle | A first-time user types the date and presses the obvious button, writing a date of birth into Last name on every row. The bulk action amplifies the mistake. |
| Recommendation | Preselect the field that matches the cause (missingDob → Date of birth, unknownTestCode → Test code, unknownUnit → Unit of the value, and so on); show the field, value and row count in the confirmation; offer "Correct and release". |
| Expected benefit | Removes the most likely bulk data-entry error in intake. |
| Complexity | Low |
| Heuristic / criterion | Nielsen 5 (error prevention) |

UX-03 **Medium** **Verified**

### Recovery codes can be skipped by navigating away

| | |
|---|---|
| Affected | First sign-in → Save your recovery codes |
| Problem | The recovery-code step shows the full navigation and header search. The session is already complete, so any menu link leaves the step without acknowledging the codes, which "will not be shown again". |
| Evidence | Observed for all eight roles: the menu and search are active on this step. Code: enrollment completes sign-in before showing the |

| | |
|---|---|
| | codes; Continue only redirects (AccountController). |
| Why users struggle | New users explore; one click on Home skips the only chance to save account-recovery codes, which leads to lockouts and help-desk calls. |
| Recommendation | Show a minimal header (brand and Sign out only) until the codes are acknowledged; or let the user regenerate recovery codes from their profile. |
| Expected benefit | Fewer lockouts; a safer first run. |
| Complexity | Low |
| Heuristic / criterion | Nielsen 5 (error prevention) |

UX-04 **Medium** **Verified**

### Lab units appear inside addresses in tasks (data defect)

| | |
|---|---|
| Affected | Work queues titles and details; address-review and investigation tasks |
| Problem | Task text reads "Investigate 410 N Milton Ave ug/dL, Baltimore, MD", and address-review details contain "Calvert ug/dL Hyattsville". A result unit has been joined into stored or derived address text. |
| Evidence | Observed in Work queues (supervisor). Root cause not established; likely in how intake builds the address from the lab file. |
| Why users struggle | Investigators rely on these addresses; corrupted text erodes trust and can misdirect field work. |
| Recommendation | Trace where task and location address text is assembled; fix the parsing; repair the affected demo records; add a test that no unit token appears in an address. |
| Expected benefit | Correct addresses in tasks. |
| Complexity | Medium |
| Heuristic / criterion | Data quality; Nielsen 2 (match the real world) |

UX-05 **Medium** **Verified**

### Home "Needs attention" repeats held rows and shows raw rule text

| | |
|---|---|
| Affected | Home dashboard |
| Problem | Each held row is listed on its own: the demo shows the same two |

| | |
|---|---|
| | rows four times each, from four uploads of one file. A task detail reads "12.0000 is at or above 10.000000 (tier below 20.000000)". The tiles "Elevated results held at intake: 8" and "Held intake records: 18" sit side by side with no explanation of the difference. |
| Evidence | Observed for the epidemiologist and the supervisor. |
| Why users struggle | Real tasks are pushed down by repeats; internal rule text and two differing held counts make the user decode the system. |
| Recommendation | Group held items by cause, with a count and one link to Held records; format thresholds as "12.0 µg/dL is at or above 10 µg/dL"; label the tiles ("Held: elevated results" vs "Held: all records"). |
| Expected benefit | A shorter, plain-language start to the day. |
| Complexity | Medium |
| Heuristic / criterion | Nielsen 2 (match the real world), 8 (minimalist design) |

UX-06 **Medium** **Verified**

## Case page shows every form at once, including high-stakes actions

| | |
|---|---|
| Affected | Case page |
| Problem | Summary editing, activity logging, Override classification and a red Close case are all expanded on one long page. The main Save is a secondary outline button while Log activity is primary, and the activity form sits above the history. |
| Evidence | Observed: page height about 2,600 px at 1280 px; screenshot in C7. |
| Why users struggle | Too many simultaneous choices; rarely used, consequential actions sit beside routine ones; primary and secondary actions are unclear. |
| Recommendation | Put Override and Close behind "Change classification…" and "Close case…" buttons that open a dialog explaining the consequences; make Save primary; show the activity history first, with "Log an activity" opening the form. |
| Expected benefit | A calmer page, fewer accidental changes, a clearer next step. |
| Complexity | Medium |
| Heuristic / criterion | Nielsen 8 (minimalist design), 5 (error prevention) |

UX-07 **Medium** **Verified**

### Work queues is difficult on a phone

| | |
|---|---|
| Affected | Work queues at 390 px |
| Problem | Five full-width bulk-action controls and the dismissal field (about 450 px) come before the first task. The table shows only Task, Queue and Person, and each row is padded with blank space set by off-screen columns. |
| Evidence | Observed at 390 px (supervisor); screenshot in C6. |
| Why users struggle | Phone users scroll past controls they can't use yet, and still can't read the task. |
| Recommendation | Show bulk actions only once something is selected (a sticky bar); show tasks as cards with title, detail, due date and person on phones. |
| Expected benefit | Usable triage in the field. |
| Complexity | Medium |
| Heuristic / criterion | WCAG 1.4.10 is met (no page overflow); Nielsen 8 |

UX-08 **Medium** **Verified**

### Location is not shown on several screens; Audit log is missing from the menu

| | |
|---|---|
| Affected | Primary navigation |
| Problem | No menu item is marked current on a lab result page, the person timeline, Jurisdiction settings, Analytes or the Audit log. Supervisors may open the Audit log but it isn't in their menu; they reach it only through "Audit log for this…" links on records. |
| Evidence | Page sweep: the current marker was empty on /results/{id}, /app/people/{id}/timeline, /config/settings, /config/analytes and /admin/audit. |
| Why users struggle | Users lose their bearings after following a link, and the audit feature is hard to discover. |
| Recommendation | Mark People for /people/* and the timeline, and Results for /results/*; mark Administration for the settings screens; add Audit log under Administration for anyone with audit.view. |
| Expected benefit | Users always know where they are and can find the audit log. |
| Complexity | Low |
| Heuristic / criterion | WCAG 2.4.8 Location (AAA), 3.2.3 Consistent Navigation; Nielsen 1 |

UX-09 **Medium** **Verified**

### Angular screens opened without permission show a vague error

| | |
|---|---|
| Affected | Held records, Match review (and other Angular screens) through a direct link |
| Problem | A case manager opening /app/held sees the full screen with "The request could not be completed." and "Nothing is held."; the server-rendered screens show a clear "Not permitted" page instead. |
| Evidence | Observed for cmbalt: HTTP 200 page, API calls 403. |
| Why users struggle | The user can't tell a permission problem from an empty queue or an outage. |
| Recommendation | Guard Angular routes with the permissions from /api/v1/me and show the same "Not permitted" page; distinguish a 403 from an empty result. |
| Expected benefit | Accurate feedback from bookmarks and shared links. |
| Complexity | Low |
| Heuristic / criterion | Nielsen 9 (recognize and recover from errors), 4 (consistency) |

UX-10 **Medium** **Verified**

### Add a person makes the user retype the search and has no h1

| | |
|---|---|
| Affected | People → Add a person |
| Problem | After a no-match search, Add a person opens an empty form: the searched name and date of birth must be retyped. The page's title "Add a person" is an h2; there is no h1. |
| Evidence | Observed: Last name empty after searching; h1 absent on /people/new (all roles). |
| Why users struggle | Retyping invites inconsistent spellings, which defeats the search-first duplicate check; screen-reader users get no page heading. |
| Recommendation | Prefill the last name, first name and date of birth from the query; make "Add a person" the h1. |
| Expected benefit | Faster, more consistent new records. |
| Complexity | Low |
| Heuristic / criterion | Nielsen 7 (efficiency); WCAG 1.3.1, 2.4.6 (best practice) |

UX-11 **Medium** **Code-inferred**

### Creating a user leaves them with no access until a second step

| | |
|---|---|
| Affected | Administration → Users → Add a user |
| Problem | The Add a user form has no role or access fields; Save redirects to the edit page, where access is added separately. Nothing on the form says so. |
| Evidence | Form observed (admin); flow read in ConfigController (SaveUser redirects to EditUser; assignments are a separate POST). Not executed: it would create an account. |
| Why users struggle | A first-time administrator may stop after Save, and the new user then signs in to an empty MERIT. |
| Recommendation | Add a "First access" section (role, tenant, level) to the create form, or show "Next: give access" after Save. |
| Expected benefit | New accounts work the first time. |
| Complexity | Low |
| Heuristic / criterion | Nielsen 1 (visibility of system status), 6 |

UX-12 **Medium** **Verified**

### Extracts: Mark submitted is the primary action and identical runs aren't told apart

| | |
|---|---|
| Affected | Registry & reports → Federal extracts |
| Problem | Each run card's primary blue button is Mark submitted, which records an external submission; Download is secondary. Three CBLS Q3 runs show the same title and file name, with nothing marking the latest. |
| Evidence | Observed (supervisor); screenshot in C8. |
| Why users struggle | Users may mark the wrong run as submitted, or download an older one. |
| Recommendation | Make Download primary; put Mark submitted behind a confirmation that names the run; badge the latest run per period and mark older ones "Superseded". |
| Expected benefit | Fewer wrong federal submissions. |
| Complexity | Low |
| Heuristic / criterion | Nielsen 5 (error prevention), 4 (consistency) |

UX-13 **Low** **Verified**

### Result entry offers three spellings of the same unit

| | |
|---|---|
| Affected | Enter a result → Unit |
| Problem | The unit list offers ug/dL, mcg/dL and µg/dL. |
| Evidence | Observed for Lead, whole blood. |
| Why users struggle | Users wonder whether they differ, and pick inconsistently. |
| Recommendation | Show one label per unit (µg/dL) and accept the other spellings as synonyms when files arrive. |
| Expected benefit | A simpler choice. |
| Complexity | Low |
| Heuristic / criterion | Nielsen 4 (consistency) |

UX-14 **Low** **Verified**

### Remaining raw codes and inconsistent formatting

| | |
|---|---|
| Affected | Person results table, result chart notes, reports, counts |
| Problem | "PB" instead of "Lead" in the person's results table; ISO dates (2026-01-01) in the chart notes beside "Oct 25, 2026" elsewhere; "task(s)" and "row(s)"; population codes in lowercase in reports; fractional axis ticks (1.3, 2.5) for whole-number counts. |
| Evidence | Observed on the person page and Reports. |
| Why users struggle | Small inconsistencies make the product feel unfinished and slow reading. |
| Recommendation | Use the existing display-name and date helpers everywhere; proper singular and plural; whole-number ticks for counts. |
| Expected benefit | Consistent reading. |
| Complexity | Low |
| Heuristic / criterion | Nielsen 4 (consistency), 2 |

UX-15 **Low** **Verified**

### Intake page uses developer language and hides the file format

| | |
|---|---|
| Affected | Intake → Upload a file |
| Problem | The help reads "Samples are in sourceCode/samples" and uses "ORU^R01" and "HL7 v2.3.1/v2.5.1" unexplained. The sender-and- |

| | |
|---|---|
| | profile list is truncated at "Chesapeake Reference Laboratory — Ch", hiding whether CSV or HL7 is chosen, and each option repeats the lab name. |
| Evidence | Observed (supervisor and intake clerk); screenshot in D. |
| Why users struggle | Clerks see internal paths and acronyms, and may pick the wrong format. |
| Recommendation | Remove the repository path; say "HL7 laboratory message (ORU)"; widen the list or split sender and format into two choices; drop the repeated name. |
| Expected benefit | A confident first upload. |
| Complexity | Low |
| Heuristic / criterion | Nielsen 2 (match the real world) |

UX-16 **Low** **Verified**

### Match review repeats the record and gives no primary choice

| | |
|---|---|
| Affected | Intake → Match review |
| Problem | With no candidates, the incoming record appears twice (summary, then the full field list). Create as a new person and Not sure are both secondary buttons, and nothing advises what to do with a placeholder name such as "Baby Boy". The filter "Sent to a supervisor only" isn't labelled as a filter. |
| Evidence | Observed (supervisor). |
| Why users struggle | The decision with the biggest consequence gets the least guidance. |
| Recommendation | Show the record once; add one line of guidance per situation (for example "Placeholder names: create a new person only if no candidate fits"); label the filter "Show only records sent to a supervisor". |
| Expected benefit | Faster, more consistent match decisions. |
| Complexity | Low |
| Heuristic / criterion | Nielsen 8, 10 (help) |

UX-17 **Low** **Verified**

### Empty link in the Work queues Person column

| | |
|---|---|
| Affected | Work queues (county case manager) |

| | |
|---|---|
| Problem | When a task's person isn't visible to the user, the Person cell still renders a link with no text. |
| Evidence | axe: link-name (serious), 1 node, tr[data-index="1"] .cdk-column-person > .usa-link (cmbalt). |
| Why users struggle | Screen-reader users hear an unnamed link; keyboard users tab to an invisible stop. |
| Recommendation | Render no link (or plain "Not in your access") when there is no person name. |
| Expected benefit | Compliant, predictable tabbing. |
| Complexity | Low |
| Heuristic / criterion | WCAG 2.4.4 Link Purpose, 4.1.2 Name, Role, Value |

UX-18 **Low** **Potential**

### Look-alike person records aren't flagged in search

| | |
|---|---|
| Affected | People search results |
| Problem | "Rivera, Mia" and "Rivera, Mya" share a date of birth, and an "Ana Rivera" with no date of birth sits beside a complete one, with no "possible duplicate" hint. |
| Evidence | Observed in the demo data; whether these are true duplicates needs program review. |
| Why users struggle | The search-first workflow depends on users spotting duplicates; a hint would help. |
| Recommendation | Validate with program staff; if useful, show a "possible duplicate of…" note using the existing match keys. |
| Expected benefit | Fewer duplicate records. |
| Complexity | Medium |
| Heuristic / criterion | Nielsen 5 |

UX-19 **Low** **Verified**

### Session-ended alert: link and button touch

| | |
|---|---|
| Affected | Session expiry alert |
| Problem | "Sign in in a new tab" and Check session sit against each other with no space, and the link touches the alert's edge. |
| Evidence | Observed after clearing cookies mid-form; screenshot in C7. |

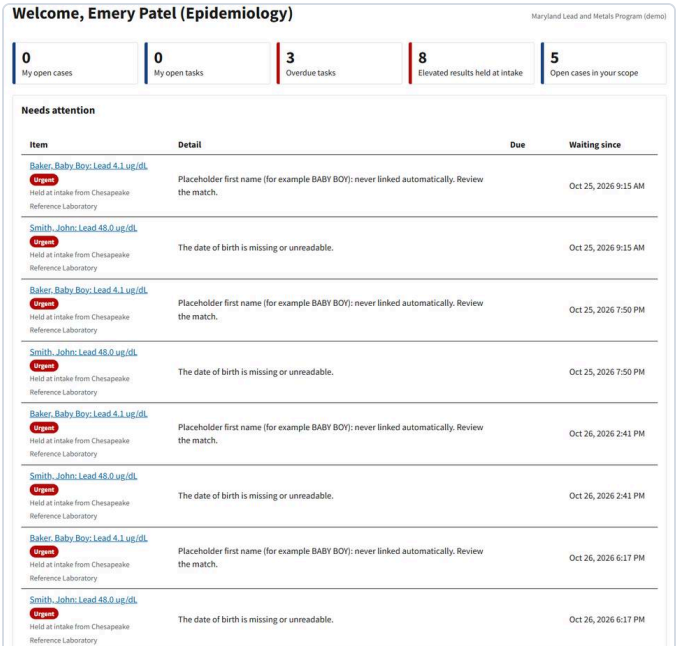
| | |
|---|---|
| Why users struggle | Small targets close together invite mis-clicks at a stressful moment. |
| Recommendation | Lay the alert's actions out as a row with a gap, inside the alert's padding. |
| Expected benefit | Clearer recovery. |
| Complexity | Low |
| Heuristic / criterion | WCAG 2.5.8 Target Size (spacing), Nielsen 8 |

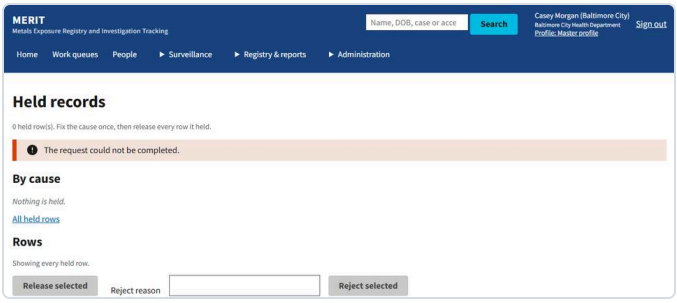
UX-20 **Low** **Verified**

### Users list filter layout glitch

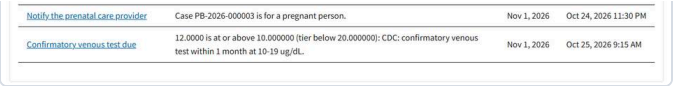
| | |
|---|---|
| Affected | Administration → Users |
| Problem | The filter's label carries an underline that looks like part of an input, and "10 users shown." is pushed against the text box. |
| Evidence | Observed (admin). |
| Why users struggle | Looks broken; the field boundary is unclear. |
| Recommendation | Use the standard label and input styles; put the count on its own line or after a gap. |
| Expected benefit | A cleaner first impression for administrators. |
| Complexity | Low |
| Heuristic / criterion | Nielsen 4 |

## Screen-level evidence




Held records opened from a link without permission (UX-09).




Home: the same held rows repeat, and one task shows raw rule text (UX-05).




Intake: a developer path in the help text, and the profile list cut off (UX-15).

---

# E. Workflow simplification recommendations

Step counts are user actions: clicks, field entries and confirmations. Fewer is not automatically better. Each proposal also removes a decision or an error opportunity.

## E1. Correct a group of held rows (UX-02)

### Current: 8 actions, 1 likely error

1. Open Held records
2. Choose the cause
3. Change *Field* from Last name to the right field
4. Type the value
5. Correct entire group
6. Confirm
7. Select the rows
8. Release selected

### Proposed: 5 actions

1. Open Held records
2. Choose the cause, with the field preset to the cause
3. Type the value
4. Correct and release the group
5. Confirm, with a summary naming the field, the value and the row count

Keep "correct only" for users who want to review before release.

## E2. Work a task from the queue (UX-01)

**Current: 6–7 actions plus scrolling**

1. Work queues
2. Scroll the grid sideways to read the detail
3. Open the person or case link
4. Go back
5. Tick the row
6. Complete
7. Confirm

**Proposed: 3–4 actions**

1. Work queues: the detail reads in full under the title
2. Open the case from the row
3. Complete from the row or the case's task list
4. Confirm

Tradeoff: rows get somewhat taller at narrow widths, so keep the detail truncated to two lines with "more".

## E3. Add a person after an unsuccessful search (UX-10)

**Current: 5 actions plus retyping**

1. Search a name and date of birth
2. No match → Add a person
3. Retype the last name
4. Retype the first name and date of birth
5. Save

**Proposed: 3 actions**

1. Search
2. Add a person, with the name and date of birth prefilled from the search
3. Check and Save

## E4. Add a user who can work (UX-11)

**Current: 2 forms, 8–9 fields, 2 saves**

1. Add a user: name, display name, email, tenant, password
2. Save
3. On the edit page, find Add access
4. Role, tenant, access level, descendants
5. Add

**Proposed: 1 form, 1 save**

1. Add a user, with a "First access" section (role, tenant, level)
2. Save, with a confirmation that names the access given

Keep the edit page for more grants.

## E5. Close a case or override its classification (UX-06)

## E3. Close a case or override its classification (UX-06)

### Current: about 4 actions, forms always visible

### Proposed: about 5 actions, but only on demand